# Companion Document: update_F.R

**q(F) — Variational Update for Genomics Loading Matrix**

Andrew Walther

2026-03-25

## Table of contents

# 1 Overview

`update_F.R` implements the CAVI variational update for the factor loading matrix **F** (shape p × K) in the joint genomics + survival model. F appears **only** in the genomics likelihood and has no role in the Cox proportional hazards survival term — this structural fact drives the module's key mathematical simplification.

The update is a **vector EBNM** (Empirical Bayes Normal Means) problem solved independently for each factor column $\mathbf{f}_k$. Within each column, the p feature-wise posterior means and variances are computed in closed form, then passed to an EBNM solver that applies the chosen prior (default: point-normal) and returns posterior moments.

**Depends on:** `compute_R_k()` from `code/update_L.R` — source that file before this one.

---

# 2 Mathematical Background

## 2.1 Model Structure

The genomics observations follow:

$$Y \approx LF^\top \quad (n \times p \text{ matrix, } L \text{ is } n \times K,\ F \text{ is } p \times K)$$

with feature-wise precision $\tau_j$ (heteroscedastic noise). The residual for factor k at feature j is:

$$R_k^{(i,j)} = Y_{ij} - \sum_{k' \neq k} \bar{L}_{ik'} \bar{F}_{jk'}$$

This is computed by `compute_R_k()` and excludes the current factor k (Gauss-Seidel schedule).

## 2.2 EBNM Sufficient Statistics for $\mathbf{f}_k$

For each feature j, the ELBO contribution from factor k leads to the following normal sufficient statistics:

| Quantity | Formula | Shape | Notes |
|---|---|---|---|
| A_F[j] | `Tau[j] * sum(EL2_k)` | p-vector | $\sum_i \mathbb{E}[l_{ik}^2]$ is a **scalar** broadcast across all j |
| B_F[j] | `Tau[j] * (t(R_k) %*% EL_k)[j]` | p-vector | Inner product of residual column with L posterior |
| x_F[j] | `B_F[j] / A_F[j]` | p-vector | EBNM observation ("sufficient statistic") |
| s_F[j] | `1 / sqrt(A_F[j])` | p-vector | EBNM noise standard deviation |

`sum(EL2_k)` is the sum of the second moments of the k-th column of L across all n samples:

$$\widetilde{L}_k^{(2)} = \sum_i \mathbb{E}[l_{ik}^2] \quad \text{(scalar; includes } \mathrm{Var}_q(l_{ik}) + \bar{l}_{ik}^2\text{)}$$

---

# 3 The Tau Cancellation Property

This is the **central mathematical insight** of the module.

## 3.1 Tau Cancels in x_j (observation)

```
x_j = B_F[j] / A_F[j]
    = (Tau[j] * (t(R_k) %*% EL_k)[j]) / (Tau[j] * sum_EL2_k)
    = (t(R_k) %*% EL_k)[j] / sum_EL2_k
```

The $\tau_j$ factors in numerator and denominator cancel exactly. The EBNM observation $x_j$ is **identical regardless of feature precision**.

## 3.2 Tau Does NOT Cancel in s_j (noise level)

```
s_j = 1 / sqrt(A_F[j]) = 1 / sqrt(Tau[j] * sum_EL2_k)
```

Here $\tau_j$ remains under the square root. A feature with higher precision (larger $\tau_j$) gets a **smaller** noise level $s_j$, which means the prior applies less shrinkage and the posterior mean stays closer to $x_j$.

### 3.3 Consequence

Features with the same residual signal $x_j$ are differentially shrunk toward zero depending on their noise precision. Clean features (high $\tau_j$) are shrunk less; noisy features (low $\tau_j$) are shrunk more. This is the heteroscedastic shrinkage mechanism for F.

| $\tau_j$ | $x_j$ | $s_j$ | Shrinkage |
|---|---|---|---|
| 1 | same | large | strong |
| 1000 | same | small (~1/sqrt(1000) smaller) | weak |

Demo 1 illustrates this with `max|x_j diff|` 0 and `s_j ratio` `sqrt(1000)`.

---

## 4 Function Reference

### 4.1 `update_F_k`

```
update_F_k(Tau, EL_k, EL2_k, R_k, prior_family = "point_normal", A_floor = 1e-10)
```

Computes the variational posterior for a single factor column $\mathbf{f}_k$.

**Arguments:**

| Parameter | Type | Description |
|---|---|---|
| `Tau` | numeric vector, length p | Feature-wise noise precisions |
| `EL_k` | numeric vector, length n | Posterior means of L column k: E_q[l_{.k}] |
| `EL2_k` | numeric vector, length n | Posterior second moments of L column k: E_q[l_{.k}^2] |
| `R_k` | numeric matrix, n × p | Residual matrix excluding factor k (from `compute_R_k`) |
| `prior_family` | character | EBNM prior family; default `"point_normal"` |
| `A_floor` | numeric | Minimum value for A_F to avoid division by zero; default `1e-10` |

**Returns:** Named list

| Field | Type | Description |
|---|---|---|
| `mean` | numeric vector, length p | Posterior mean E_q[f_{.k}] |
| `second` | numeric vector, length p | Posterior second moment E_q[f_{.k}^2] |
| `sd` | numeric vector, length p | Posterior standard deviation |
| `A` | numeric vector, length p | EBNM precision A_F (after floor) |
| `B` | numeric vector, length p | EBNM numerator B_F |
| `x` | numeric vector, length p | EBNM sufficient statistic x_F = B/A |
| `s` | numeric vector, length p | EBNM noise s_F = 1/sqrt(A) |
| `sum_EL2_k` | numeric scalar | sum(EL2_k), shared across all features |
| `ebnm_result` | list | Raw output from `ebnm()` call |

**Key implementation details:**

- `A_F = pmax(Tau * sum(EL2_k), A_floor)` — the floor prevents NaN when Tau is zero
- `B_F = Tau * (t(R_k) %*% EL_k)` — matrix-vector product gives p-vector
- Second moment satisfies: `second[j] = sd[j]^2 + mean[j]^2`

---

### 4.2 `update_F_all`

```
update_F_all(Y, EL, EL2, EF, EF2, Tau, prior_family = "point_normal", A_floor = 1e-10)
```

Gauss-Seidel loop over all K factor columns.

**Arguments:**

| Parameter | Type | Description |
|---|---|---|
| `Y` | numeric matrix, n $\times$ p | Observed genomics data |
| `EL` | numeric matrix, n $\times$ K | Posterior means of L (fully updated from L step) |
| `EL2` | numeric matrix, n $\times$ K | Posterior second moments of L |
| `EF` | numeric matrix, p $\times$ K | Current posterior means of F (updated in place) |
| `EF2` | numeric matrix, p $\times$ K | Current posterior second moments of F |

| Parameter | Type | Description |
|---|---|---|
| `Tau` | numeric vector, length p | Feature-wise precisions |
| `prior_family` | character | EBNM prior family; default `"point_normal"` |
| `A_floor` | numeric | Floor for A; default `1e-10` |

**Returns:** Named list

| Field | Type | Description |
|---|---|---|
| `EF` | numeric matrix, p $\times$ K | Updated posterior means of F |
| `EF2` | numeric matrix, p $\times$ K | Updated posterior second moments of F |
| `details` | list of length K | Per-factor output from `update_F_k` |

**Important Gauss-Seidel note:**

`update_F_all` passes the **fully updated** `EL` (not a stale copy) because the L step completes before the F step in the CAVI schedule. Within the F loop, `EF_curr` is updated column-by-column so that each `compute_R_k` call sees the most recent estimates of all other F columns. This is the Gauss-Seidel (sequential) schedule, as opposed to a Jacobi (parallel) schedule.

```
for k in 1:K:
    R_k  <- compute_R_k(Y, EL, EF_curr, k)   # excludes column k
    res  <- update_F_k(Tau, EL[,k], EL2[,k], R_k, ...)
    EF_curr[,k]  <- res$mean
    EF2_curr[,k] <- res$second
```

---

# 5 Test Suite

File: `tests/test_update_F.R` Total: **26 tests across 9 groups** — all passing.

## 5.1 T1: Mathematical Identity Checks (6 tests)

| Test | What is verified |
|---|---|
| T1.1 | `A_F[j] = Tau[j] * sum(EL2_k)` exactly |
| T1.2 | `B_F[j] = Tau[j] * (t(R_k) %*% EL_k)[j]` exactly |
| T1.3 | `x` is a p-vector (length p) |

| Test | What is verified |
|---|---|
| T1.4 | Tau cancels in x_j: `Tau=1` vs `Tau=1000` produce identical `x_j` |
| T1.5 | Tau does NOT cancel in s_j: `s_j` ratio is significantly different |
| T1.6 | Second moment identity: `second[j] = sd[j]^2 + mean[j]^2` |

### 5.2 T2: WLS / Non-Informative Limit (2 tests)

| Test | What is verified |
|---|---|
| T2.1 | When `EL2_k = EL_k^2` (zero posterior variance in L), `x_j` equals the OLS estimator |
| T2.2 | Large Tau drives posterior mean close to the observation `x_j` |

### 5.3 T3: Known Signal Recovery K=1 (3 tests)

| Test | What is verified |
|---|---|
| T3.1 | Dense factor: correlation between recovered and true f exceeds 0.7 |
| T3.2 | Sparse factor (50/500 nonzero): active features have larger posterior mean than inactive |
| T3.3 | Higher Tau $\rightarrow$ better recovery (monotone precision effect) |

### 5.4 T4: Multi-Factor K=5 (3 tests)

| Test | What is verified |
|---|---|
| T4.1 | Output matrices EF, EF2 have shape p $\times$ K |
| T4.2 | All second moments >= squared means (`EF2 >= EF^2`) component-wise |
| T4.3 | All outputs are finite (no NaN or Inf) |

### 5.5 T5: Null Factor Shrinkage (2 tests)

| Test | What is verified |
|---|---|
| T5.1 | Pure-noise R_k produces factor near zero (point-normal shrinks to zero) |
| T5.2 | Zero EL_k $\rightarrow$ A hits floor, B=0 $\rightarrow$ EF near zero |

### 5.6 T6: Tau Differential Shrinkage (3 tests)

| Test | What is verified |
|---|---|
| T6.1 | High-tau features are shrunk less than low-tau features (same x_j) |
| T6.2 | Larger EL2 → larger A → smaller s → less shrinkage for high-signal factors |
| T6.3 | Uniform Tau → all `s_j` equal (homoscedastic limit, one shared noise level) |

### 5.7 T7: Numerical Stability (4 tests)

| Test | What is verified |
|---|---|
| T7.1 | All-zero Tau → A hits floor, no NaN or Inf in output |
| T7.2 | Extreme Tau=1e10 → no overflow in A, B, x, s |
| T7.3 | Edge case p=1, n=1 → correct scalar outputs |
| T7.4 | Custom A_floor value is respected (A >= A_floor) |

### 5.8 T8: R_k and Gauss-Seidel (2 tests)

| Test | What is verified |
|---|---|
| T8.1 | R_k correctly uses EL_k (not another factor's column) |
| T8.2 | `update_F_all` produces finite results with valid second moments across all K |

### 5.9 T9: V2.R Consistency (1 test)

| Test | What is verified |
|---|---|
| T9.1 | `update_F_k` output matches the corresponding lines in `Supervised_Bayesian_MF_V2.R` (lines 334–340) exactly |

---

## 6 Demos

File: `demos/demo_update_F.R` Total: **5 scenarios** — narrative output, no PASS/FAIL.

### 6.1 Demo 1: Tau Cancellation Property

**Purpose:** Demonstrate the core mathematical property that $\tau_j$ cancels in $x_j$ but not in $s_j$.

**Setup:** Same Y, L, R_k. Run `update_F_k` twice — once with `Tau = rep(1, p)`, once with `Tau = rep(1000, p)`.

**Expected output:** - `max|x_j(Tau=1) - x_j(Tau=1000)|` 0 (machine precision) - `s_j(Tau=1) / s_j(Tau=1000)` `sqrt(1000)` `31.6` for all j

This illustrates that the EBNM "data" is precision-free, while the EBNM "noise level" encodes the precision.

---

### 6.2 Demo 2: Signal Recovery K=1

**Purpose:** Confirm that the update recovers a sparse true factor from noisy observations.

**Setup:** n=150, p=200, K=1. True f_true is sparse (30/200 nonzero entries, drawn from N(0,1)). Y = L %*% t(f_true) + noise.

**Reported metrics:** - Pearson correlation between `EF[,1]` and `f_true` - Mean posterior mean for active vs inactive features (active should be larger)

---

### 6.3 Demo 3: Tau-Dependent Shrinkage

**Purpose:** Show that heteroscedastic precision differentially regularizes features.

**Setup:** p=200 features split into two groups: first 100 with `tau=10` (clean), last 100 with `tau=0.5` (noisy). Same true signal in both groups.

**Reported metrics:** - Mean absolute error (MAE) for clean vs noisy features - Posterior SDs for clean vs noisy features - Demonstrates that clean features (high Tau) stay closer to truth

---

### 6.4 Demo 4: Sparsity Recovery

**Purpose:** Evaluate the point-normal prior's ability to identify which features belong to a sparse factor.

**Setup:** n=100, p=250, true f is sparse (40/250 nonzero). Threshold posterior means at 0.1 to classify active/inactive.

**Reported metrics (confusion matrix):**

| | Predicted Active | Predicted Inactive |
|---|---|---|
| **True Active** | TP | FN |
| **True Inactive** | FP | TN |

Also reports precision = TP/(TP+FP) and recall = TP/(TP+FN).

---

### 6.5 Demo 5: Multi-Factor K=5

**Purpose:** Validate `update_F_all` on a structured 5-factor problem.

**Setup:** n=200, p=300, K=5. Each factor has a distinct sparse structure; F is initialized at zero.

**Reported metrics:** - Column-wise correlations: `cor(EF[,k], F_true[,k])` for each k - Number of active features per factor (posterior mean > 0.1) - Confirms Gauss-Seidel within-loop updates do not break structure

---

## 7 Computational Notes

### 7.1 Scalar Broadcast

$\sum_i \mathbb{E}[l_{ik}^2]$ is computed once (a single scalar) and broadcast across all p features via element-wise multiplication with $\tau_j$. This avoids a p × n matrix multiply for $A_F$.

### 7.2 Shared Dependency: compute_R_k

`compute_R_k` is defined in `code/update_L.R`. Always source `update_L.R` before `update_F.R`. The residual matrix R_k has shape n × p and represents the data with all other factors' contributions removed. Because L is fully updated before the F step, the R_k computation here uses the most recent L posterior.

### 7.3 Memory Layout

| Matrix | Shape | Update order |
|---|---|---|
| EL | n × K | Fixed (updated in L step) |
| EL2 | n × K | Fixed (updated in L step) |
| EF | p × K | Updated column by column (Gauss-Seidel) |
| EF2 | p × K | Updated column by column |
| R_k | n × p | Recomputed fresh for each k |

| Matrix | Shape | Update order |
|---|---|---|

For large p, R_k (n $\times$ p) dominates memory. It is not cached between factor updates.

---

## 8 Related Files

| File | Relationship |
|---|---|
| `code/update_L.R` | Defines `compute_R_k()` required by this module |
| `code/update_tau.R` | Provides Tau used as input here |
| `code/update_beta.R` | Beta update (survival); F does not appear there |
| `code/Supervised_Bayesian_MF_v2.R` | Main algorithm; lines 334–340 are the reference for T9.1 |
| `tests/test_update_F.R` | 26 tests, all passing |
| `demos/demo_update_F.R` | 5 narrative demos |
| `derivations/qF/qF_update_derivation.tex/.pdf` | Full derivation including tau cancellation proof |
| `code/SupervisedMF_Context.md` | AI/code quick-reference for the full R implementation |